

Supplementary Material (v2.0.0)

A Computable Map of Treatment-Sequencing Evidence in HR+/HER2- Metastatic Breast Cancer: A 64-Trial, Pre-Registered Graph-Theoretic Analysis of ESMO, ASCO, and NCCN Decision-Tree Concordance

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Supplementary Table S1. Reporting Transparency Checklist

This study reports a pre-registered graph-theoretic computational analysis. We adopt a custom transparency checklist combining PRISMA-2020 reporting items (relevant to evidence synthesis) with a graph-encoding addendum specific to the present method.

Item	Reported in	Notes
1. Title identifies evidence synthesis	Title	“Computable Map of Treatment-Sequencing Evidence”
2. Structured abstract	Abstract	268-word JCO CCI structure
3. Rationale	Introduction	Composition across non-overlapping trials
4. Objectives	Introduction	EFDPR + ODI
5. Eligibility criteria for trials	Methods 2.1, 2.2	Eight pre-specified filters (<code>v2_02_filter_corpus.py</code>)
6. Information sources	Methods 2.1	ClinicalTrials.gov v2 + ESMO 2024 + ASCO 2022 + NCCN 2024
7. Search strategy	Methods 2.2	Frozen API query at <code>v2_01_systematic_search.py</code>
8. Selection process	Methods 2.2	Eight pre-specified filters; 874 candidates → 82 trials
9. Data extraction process	Methods 2.3	Dual-annotator LLM (Claude + Codex/GPT-5) + protocol adjudication
10. Data items	Methods 2.3	Frozen schema v1.0 fields
11. Risk-of-bias assessment	Discussion	Selection bias disclosed; multi-round internal review log
12. Effect measures	Methods 2.7	EFDPR; ODI
13. Synthesis methods	Methods 2.6	Concordance algorithm; tolerance grid
14. Reporting bias assessment	Limitations	Snapshot date; corpus expansion disclosed
15. Certainty assessment	Discussion	Pre-registered tolerance levels; sensitivity by guideline source
16. Study selection results	Results 3.2	74 systematic + 6 supplementary + 2 round-1-added = 82

Item	Reported in		Notes
17. Study characteristics	Table 1 + S3		Per-trial schema fills
18. Risk of bias	Limitations		Disclosed
19. Results of individual studies	DAG edges		Trial primary endpoint per edge
20. Results of syntheses	Results 3.3–3.6		EFDPR, ODI, sensitivity, secondary outcomes
21. Reporting biases	Discussion		Reviewer-cycle history disclosed
22. Certainty of evidence	Discussion		Pre-registered exact test rejected at $P = 0.011$
23. Discussion	Discussion		Three caveats + three positives + outlook
24. Limitations	Discussion		Pilot-to-production scaling honestly framed
25. Implications	Discussion		Outlook
26. Registration and protocol	Methods	2.1, prereg-v2	docs/prereg-v2.md at commit 085ae54 + amendment 079b540
27. Support	Funding	statement	No specific funding
28. Competing interests	Conflicts	statement	None declared
29. Availability of data, code, materials	Statements		MIT licence, GitHub URL, Zenodo DOI at v2.0.0
G1. Graph node space	Methods 2.4		(prior-state, biomarker) tuples
G2. Graph edge construction	Methods 2.4		Trial $\rightarrow$ edge from required-state node
G3. Guideline-tree encoding	Methods 2.5		25-node ESMO+ASCO+NCCN unified
G4. Concordance algorithm	Methods 2.6		State-superset + biomarker tolerance + drug match + temporal precedence
G5. Tolerance levels	Methods 2.6		Strict / ESCAT / liberal, pre-registered sensitivity grid
G6. Reproducibility (seed + commit hash)	Methods 2.8		Seed 20260516; v2.0.0 commit

Supplementary Table S2. Inter-annotator Agreement (Cohen’s κ and PABAK)

Cohen’s κ and the prevalence-adjusted bias-adjusted kappa (PABAK = $2P_0 - 1$) on the 20-trial validation subset (pre- and post-adjudication). Adjudication rules (17) applied per the protocol’s decision-rule hierarchy.

Field	Pre-adj κ	Post-adj κ	Post-adj agreement	PABAK	Gate ≥ 0.70
prior_state.post_endo	0.398	1.000	100%	1.000	Pass (Cohen + PABAK)
prior_state.post_cdk46i	0.667	1.000	100%	1.000	Pass (Cohen + PABAK)
biomarker.hr_pos	0.643	1.000	100%	1.000	Pass (Cohen + PABAK)
biomarker.her2_neg	0.649	1.000	100%	1.000	Pass (Cohen + PABAK)
biomarker.her2_low	1.000	1.000	100%	1.000	Pass (Cohen + PABAK)
biomarker.pik3ca_mut	1.000	1.000	100%	1.000	Pass (Cohen + PABAK)
biomarker.esr1_mut	1.000	1.000	100%	1.000	Pass (Cohen + PABAK)
biomarker.akt_path	0.000	0.000	95%	0.900	Fail Cohen, Pass PABA

Note on the kappa paradox for akt_path. When two annotators agree on a category that is

rare in the marginal distribution (both annotators rate the field as `null` on 19 of 20 trials), Cohen’s κ is suppressed despite high raw agreement. This is the well-documented kappa paradox; PABAK is the standard adjustment. We report both metrics and explicitly disclose that the pre-registered Cohen’s $\kappa \geq 0.70$ gate fails on `akt_path` alone (due to the paradox, not substantive disagreement); under PABAK the gate passes on every key field.

Supplementary Text 1. Pre-registration trail and v2 deviations

The v2 pre-registration is in two commits:

- `docs/prereg-v2.md` at commit `085ae54` (initial freeze, before any v2 outcome data observed).
- Amendment v2.1 at commit `079b540` (widened date filter to 2013–2026; tightened enrolment ≥ 200 ; disclosed before any LLM extraction).

v2 deviations disclosed:

- **Two trials added in round-1 internal review (BYLieve NCT03056755, SERENA-6 NCT04964934).** The systematic search and supplementary inclusion produced 80 trials; internal review surfaced two pivotal trials missing from the corpus that are guideline-cited and clinically pivotal. Added before primary-test computation was finalised; the addition is documented in the repository commit log.
- **Three extraction errors corrected in round-1.** CAPItello-291 and DESTINY-Breast06 post-CDK4/6i subgroup-readout flags were missing; TROPION-Breast01 HER2-low subgroup-readout flag was missing. All three corrections were applied to the final extraction file with explicit rationale.
- **17 adjudication rules (not 14 as initially reported)** were applied per the protocol’s decision-rule hierarchy.
- **Cohen’s $\kappa \geq 0.70$ gate failed pre-adjudication** on `post_endo` ($\kappa = 0.40$) and `akt_path` ($\kappa = 0.00$); post-adjudication all fields pass under PABAK but `akt_path` still fails Cohen’s κ due to the kappa paradox. Both metrics reported.
- **Pre-registered S2 and S3 reported descriptively** rather than with formal hypothesis tests, due to small corpus and the realistic null distribution for temporal lag.
- **By-guideline-source sensitivity (E1)** promoted from exploratory to pre-specified sensitivity reporting; not re-classified as confirmatory.
- **G6 (AKT-pathway) state coding** retained as `post_endo` (per v1.0.0 Round 4 review); not re-deviating.
- **NCT-label correction.** An earlier internal draft confused NCT04711252 (SERENA-4, first-line) with NCT04964934 (SERENA-6, ctDNA-detected ESR1mut switch); corrected to the latter.

Supplementary Table S3. Trial Corpus (provided as JSON in repository)

The full 82-trial structured extraction (post-adjudication) is released as `data/processed/v2_extraction_final.json` in the repository; the per-trial fields include `nct_id`, `trial_name`, `prior_state`, `biomarker`, `subgroup_readouts`, `drug_class`, `year_pc`, `guideline_target_node`, `confidence`, `notes`, and `provenance` (`systematic_search`, `v1_supplementary`, or `v2_round1_added`).

Supplementary Table S4. Guideline Decision Tree Encoding

The 25-node ESMO+ASCO+NCCN HR+/HER2- mBC decision-tree encoding is released as `data/processed/v2_` with each node carrying (node_id, state, biomarker, recommended_classes with grade, cited_trials, year, source).

Supplementary Table S5. Operationalization Discordance Tokens

Biomarker-token vocabulary for the ODI computation is in `analysis/v2_10_compute_odi.py`.

Supplementary Table S6. Reviewer-Cycle History

Four-round v1 review cycle (16 review passes) is in `reviews/round[1-4]/`; one-round v2 review cycle (4 review passes) is in `reviews/v2_round1/`; integration commits are in the repository git log.